Rationalising the Denominator

AQA · KS4 Year 10–11

Answer each question in your workbook. Show your working.

COMMON MISTAKES

- Must multiply BOTH top and bottom by the same thing.
- For (a + b) use conjugate (a - b), not just b.
- (3)^2 = 3, not 3.
- Don't forget to simplify at the end.

METHOD

1. Simplify: multiply top and bottom by a
2. a × a = a denominator rational
3. Complex: multiply by the conjugate (a b)
4. (a+ b)(a- b) = a^2 - b
5. Simplify the fraction

WORKED EXAMPLE

Rationalise 5/(3 + 2).
Conjugate: (3 - 2)
Top: 5(3 - 2) = 15 - 5 2
Bottom: (3+ 2)(3- 2) = 9 - 2 = 7
Answer: (15 - 5 2)/7

VISUAL AID

Decision tree: Is the denominator just a? Yes multiply by a/ a. No Is it (a ± b)? Yes multiply by conjugate (a b)/(a b).

1. (2 marks)

Rationalise: a) 1/ 3 b) 1/ 5 c) 1/ 7 d) 1/ 2

2. (2 marks)

Rationalise: a) 4/ 2 b) 6/ 3 c) 10/ 5 d) 3/ 6

3. (3 marks)

Rationalise and simplify: a) 12/ 6 b)   2/ 8 c)   5/ 20 d) (2 3)/ 12

4. (3 marks)

Rationalise: a) 1/(1+ 2) b) 1/(3- 5) c) 2/(4+ 3)

5. (3 marks)

Rationalise and simplify: a) 6/(3+1) b) 10/(5- 3) c) (1+ 2)/(1- 2)

6. (4 marks)

Rationalise: a) (3+ 5)/(2- 5) b) (7+2)/(7-2)

7. (4 marks)

a) Show 1/(5-2) - 1/(5+2) = 4. b) Express (2+ 3)/(2- 3) as a+b 3.

8. (5 marks)

a) Rationalise (3+ 2)/(3- 2) fully. b) Rectangle area 6 cm^2, width (3-1). Exact length?